Convex optimization for machine learning – TD5

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1 Exercises

Exercise 1. (Support vector machine)

Consider a training set formed by couples $(\mathbf{x}_i, y_i)$ for $i \in \{1, \dots, n\}$ where $\mathbf{x}_i$ is a feature vector in $\mathcal{X}$ and $y_i \in \{-1, +1\}$ for all i. The (affine) hyperplane $\mathcal{H}_{\mathbf{w},b} := \{\mathbf{x} \in \mathbb{R}^d,\ \langle \mathbf{w}, \mathbf{x}\rangle + b = 0\}$ is called *separating* if

$$\forall i, \quad y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) > 0\,.$$

In the sequel, we assume that a separating hyperplane exists. Among all separating hyperplanes, we seek to find the one which maximizes the minimum distance

$$f(\mathbf{w}, b) = \min_{i=1,\dots,n} d(\mathbf{x}_i, \mathcal{H}_{\mathbf{w},b})\,.$$

1. Show that if $(\mathbf{w}, b)$ defines a separating hyperplane, then[1] $f(\mathbf{w}, b) = c(\mathbf{w}, b)/\|\mathbf{w}\|$ where $c(\mathbf{w}, b) = \min_i y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b)$.

Thus, we are interested in solving the problem

$$\max_{\mathbf{w},b} \frac{c(\mathbf{w}, b)}{\|\mathbf{w}\|} \text{ such that } \forall i,\ y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) \geq 0\,. \tag{1}$$

Let $(\mathbf{w}^*, b^*)$ be a solution and define

$$\mathbf{v}^* = \frac{\mathbf{w}^*}{c(\mathbf{w}^*, b^*)} \text{ and } \quad a^* = \frac{b^*}{c(\mathbf{w}^*, b^*)}$$

2. Justify that $(\mathbf{w}^*, b^*)$ and $(\mathbf{v}^*, a^*)$ define the same separating hyperplane.

3. Prove that $(\mathbf{v}^*, a^*)$ solves the optimization problem

$$\max_{\mathbf{v},a} \frac{1}{\|\mathbf{v}\|} \text{ such that } \forall i,\ y_i(\langle \mathbf{v}, \mathbf{x}_i\rangle + a) \geq 1\,.$$

4. Deduce that $(\mathbf{v}^*, a^*)$ solves the optimization problem

$$\min_{\mathbf{v},a} \frac{\|\mathbf{v}\|^2}{2} \text{ such that } \forall i,\ 1 - y_i(\langle \mathbf{v}, \mathbf{x}_i\rangle + a) \leq 0\,. \tag{2}$$

What is the purpose of this question? (in comparison to the previous one for example)

5. Write the Lagrangian[2] $L(\mathbf{v}, a; \phi)$.

[1] Remember the previous tutorial where we proved (or not) that:

$$\mathrm{d}(\mathbf{z}, \mathcal{H}_{\mathbf{w},b}) = \min_{\mathbf{x} \in \mathcal{H}_{\mathbf{w},b}} \|\mathbf{x} - \mathbf{z}\| = \frac{|\langle \mathbf{w}, \mathbf{z}\rangle + b|}{\|\mathbf{w}\|}\,.$$

[2] Notice the ';' which separates the primal variables $(\mathbf{v}, a)$ from the dual variable ϕ.

6. Write the KKT conditions.

7. Let $(\mathbf{v}, a; \phi)$ be a saddle point of the Lagrangian. Show that ϕ_i is non-zero only if $y_i(\langle \mathbf{v}, \mathbf{x}_i \rangle + a) = 1$.

The training points $(\mathbf{x}_i, y_i)$ satisfying the above property are the closest to the hyperplane $\mathcal{H}_{\mathbf{v},a}$. The corresponding $\mathbf{x}_i$'s are often called *support vectors*.

8. If one is given a dual solution $\phi^\star$, how to recover a primal solution $(\mathbf{v}^*, a^*)$ from $\phi^\star$?

Define the $n \times n$ matrices $K = (\langle \mathbf{x}_i, \mathbf{x}_j \rangle)_{i,j=1\ldots n}$, $D = \mathrm{diag}(y_1 \ldots y_n)$ and $\mathbf{1}^T = (1, \ldots, 1)$.

9. Prove that the dual problem reduces to

$$\min_{\substack{\phi \geq 0 \\ \langle \mathbf{y}, \phi \rangle = 0}} \frac{1}{2} \langle KD\phi, D\phi \rangle - \langle \mathbf{1}, \phi \rangle .$$

10. Assume that this algorithm has identified a dual solution $\phi^\star$. Write explicitely the classifier as a function of $\phi^\star$.

11. What part of the training data do you need in order to implement the above classifier?

2 Solutions

Exercise 1 1. Using the recalled identity in the footnote, we get that:

$$f(\mathbf{w}, b) = \min_{i=1,\dots,n} \frac{|\langle \mathbf{w}, \mathbf{x}_i \rangle + b|}{\|\mathbf{w}\|} = \min_{i=1,\dots,n} \frac{|y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b)|}{\|\mathbf{w}\|} = \min_{i=1,\dots,n} y_i \frac{\langle \mathbf{w}, \mathbf{x}_i \rangle + b}{\|\mathbf{w}\|},$$

since we know that all $y_i\langle \mathbf{w}, \mathbf{x}_i \rangle + b$ are positive and $y_i = \pm 1$.

2. We have to prove that $\mathcal{H}_{\mathbf{v}^*,a^*} = \mathcal{H}_{\mathbf{w}^*,b^*}$. But if $\mathbf{x} \in \mathcal{H}_{\mathbf{w}^*,b^*}$, then:

$$\langle \mathbf{v}^*, \mathbf{x} \rangle + a^* = \frac{\langle \mathbf{w}^*, \mathbf{x} \rangle}{c(\mathbf{w}^*, b^*)} + \frac{b^*}{c(\mathbf{w}^*, b^*)} = 0$$

by definition of $\mathcal{H}_{\mathbf{w}^*,b^*}$, and so $\mathcal{H}_{\mathbf{w}^*,b^*} \subset \mathcal{H}_{\mathbf{v}^*,a^*}$. The reciprocal is obvious, so we get the desired result.

3. Since $(\mathbf{w}^*, b^*)$ is a solution of (1), and:

$$\frac{c(\mathbf{w}, b)}{\|\mathbf{w}\|} = \frac{1}{\left\| \frac{\mathbf{w}}{c(\mathbf{w},b)} \right\|},$$

we see that replacing $\mathbf{w}^*$ by $\mathbf{v}^* = \mathbf{w}^*/c(\mathbf{w}^*, b^*)$ and b^* by $a^* = b^*/c(\mathbf{w}^*, b^*)$ yields that $(\mathbf{v}^*, a^*)$ satisfies:

$$\frac{1}{\|\mathbf{v}^*\|} = \max_{\mathbf{v},a} \frac{1}{\|\mathbf{v}\|},$$

with the constraint that

$$\forall j,\ y_j(\langle \mathbf{v}^*, \mathbf{x}_j \rangle + a^*) = \frac{1}{c(\mathbf{w}^*, b^*)} y_j(\langle \mathbf{w}^*, \mathbf{x}_j \rangle + b^*) \geq \frac{1}{c(\mathbf{w}^*, b^*)} \min_i\ y_j(\langle \mathbf{w}^*, \mathbf{x}_j \rangle + b^*) = 1.$$

4. It suffices to notice that $h : x \mapsto 1/(2x^2)$ is a non-increasing function which turns $1/\|\mathbf{v}\|$ in $\|\mathbf{v}\|^2/2$, and the maximization problem into a minimization one:

$$\min_{\mathbf{v},a} \frac{\|\mathbf{v}\|^2}{2} = h\Big(\max_{\mathbf{v},a} \frac{1}{\|\mathbf{v}\|}\Big)$$

The constraint does not change. The main purpose of this question is to replace the non-differentiable function $\mathbf{v} \mapsto 1/\|\mathbf{v}\|$ by $\mathbf{v} \mapsto \|\mathbf{v}\|^2/2$, which will be much more pleasant to work with. Notice also that normalizing by $c(\mathbf{w}^*, b^*)$ allows us to replace the strict inequalities $y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + b) \geq 0$ by $y_i(\langle \mathbf{w}, \mathbf{x}_i \rangle + a) \geq 1$.

5. With the notations of the lecture, the function f is $\mathbf{v} \mapsto \|\mathbf{v}\|^2/2$ while g is:

$$\mathbf{x} \mapsto \big(g_1(\mathbf{x}), \dots, g_n(\mathbf{x})\big), \text{ where } g_i(\mathbf{x}) = 1 - y_i(\langle \mathbf{v}, \mathbf{x}_i \rangle + a),$$

and the associated Laplacian equals:

$$L(\mathbf{v}, a; \phi) = \frac{1}{2}\|\mathbf{v}\|^2 + \langle \phi, g(\mathbf{x}) \rangle - \iota_{\mathbb{R}^n_+}(\phi).$$

6. There are four conditions which together are known as the KKT conditions and must be satisfied by $(\mathbf{v}^*, a^*, \phi^*)$ to be a saddle point of the Lagrangian. First we give them as they are formulated in the lecture notes (see theorem 6.3.2. page 35), and then we rewrite them in a more useful way. Let $\mathbf{y} := (y_1, \dots, y_n)$. Remember that in the course x denotes the primal variable(s). Here $x = (\mathbf{v}, a)$. Also remember that for fixed ϕ such that $\phi_i \leq 0$ for all i (and so $\iota_{\mathbb{R}^n_-}(\phi) = 0$), we know that $L(\cdot, \cdot; \phi)$ is differentiable, so we can use the usual gradient instead of the subdifferential.

   - *Primal feasibility:* $g(\mathbf{v}^*, a^*) \preceq \mathbf{0}_{\mathbf{R}^n}$, that is, $\forall i \in [\![1, n]\!],\ g_i(\mathbf{v}^*, a^*) \leq 0$.
   - *Dual feasibility:* $\phi^* \succeq \mathbf{0}_{\mathbb{R}^n}$, that is, $\forall i \in [\![1, n]\!],\ \phi_i^* \leq 0$.
   - *Complementary slackness:* $\langle \phi^*, g(\mathbf{v}^*, a^*) \rangle = 0$.
   - *First order primal feasibility:*

$$\nabla_{(\mathbf{v},a)} L(\mathbf{v}^*, a^*; \phi^*) = \mathbf{0}_{\mathbb{R}^{n+1}}.$$

   In the sequel, it will be more convenient to work with those expressions, which are exactly the previous conditions, but formulated differently:

   - *Primal feasibility:* $\forall i \in [\![1, n]\!],\ y_i(\langle \mathbf{v}^*, \mathbf{x}_i \rangle + a^*) \geq 1$.
   - *Dual feasibility:* $\forall i \in [\![1, n]\!],\ \phi_i^* \geq 0$.
   - *Complementary slackness:* $\forall i \in [\![1, n]\!],\ g_i(\mathbf{v}^*) = 1 - y_i(\langle \mathbf{v}^*, \mathbf{x}_i \rangle + a^*) = 0$ or $\phi_i^* = 0$.
   - *First order primal feasibility:*

$$\nabla_{\mathbf{v}} L(\mathbf{v}^*, a^*; \phi^*) = \mathbf{v}^* - \sum_{i=1}^{n} y_i \phi_i^* \mathbf{x}_i^* = \mathbf{0}_{\mathbb{R}^n},$$

$$\partial_a L(\mathbf{v}^*, a^*; \phi^*) = -\langle \mathbf{y}, \phi^* \rangle = 0.$$

   The notation $\nabla_{(\mathbf{v},a)} L(\mathbf{v}, a; \phi)$ denotes the gradient of the Lagrangian with respect to $\mathbf{v}$ and a. It outputs a vector of $\mathbb{R}^{n+1}$ which formally reads as:

$$\nabla_{(\mathbf{v},a)} L(\mathbf{v}, a; \phi) = \begin{pmatrix} \nabla_{\mathbf{v}} L(\mathbf{v}, a, \phi) \\ \partial_a L(\mathbf{v}, a) \end{pmatrix}$$

   The reformulated third condition is a consequence of the original complementary slackness condition, the original primal feasibility condition, as well as of the dual feasibility condition: we know that $\langle \phi^*, g(\mathbf{v}^*, a^*) \rangle = 0$, that $\forall i \in [\![1, n]\!],\ g_i(\mathbf{v}^*, a^*) \leq 0$ and $\phi_i \leq 0$. So we get that a sum of nonnegative terms vanishes. Thus all terms must be equal to zero, which gives the reformulated complementary slackness condition.

7. This is a direct consequence of the complementary slackness condition: $\phi^* \neq 0$ so

$$\forall i \in [\![1, n]\!],\ y_i(\langle \mathbf{v}^*, \mathbf{x}_i \rangle + a^*) = 1.$$

8. Let us explain what this question means exactly: assume that you want to solve a complicated optimization problem. Maybe the dual problem is easier, so you manage to find its solution ϕ^*. Now you want to deduce from that $\mathbf{x}^* = (\mathbf{v}^*, a^*)$ the solution of your original problem, the primal problem. But you know that $(\mathbf{v}^*, a^*; \phi^*)$ is a saddle point of your Lagrangian. So KKT conditions will help you express the unknown $(\mathbf{v}^*, a^*)$ in terms of what you know, that is, ϕ^*.

   The first equation of the first order primal feasibility condition yields:

$$\mathbf{v}^* = \sum_{i=1}^{n} y_i \phi_i^* \mathbf{x}_i^*$$

(and the right-hand side does not depend on $\mathbf{v}^*$ nor a^*!). And using the previous question and the fact that $y_i^2 = 1$, we can recover a^* as well:

$$\forall i \in [\![1, n]\!] \text{ such that } \phi_i^* \neq 0,\ a^* = y_i - \langle \mathbf{v}^*, \mathbf{x}_i \rangle,$$

If this condition is never satisfied, that is, if all $\phi_i^* = 0$, then $\mathbf{v}^* = \mathbf{0}_{\mathbb{R}^n}$ and by the primal feasibility condition we have:

$$\forall i \in [\![1, n]\!],\ y_i a^* \geq 1,$$

which is possible only if every $y_i = 1$ or $y_i = -1$ for all i. Thus as long the two classes of the classification problem have at least one element each, we can find one index i_0 such that $\phi_{i_0} \neq 0$. As a result, using the previous expression of $\mathbf{v}^*$, we can take:

$$a^* = y_{i_0} - \langle \mathbf{v}^*, \mathbf{x}_{i_0} \rangle = y_{i_0} - \sum_{i=1}^{n} y_i \phi_i^* \langle \mathbf{x}_i, \mathbf{x}_{i_0}^* \rangle.$$

9. The dual problem is found by keeping the dual variables feasible[3] and minimizing the Lagrangian with respect to the primal variables. As before, the first-order conditions are:

$$\nabla_v L(\mathbf{v}^*, a^*; \phi) = \mathbf{v}^* - \sum_{i=1}^{n} y_i \phi_i \mathbf{x}_i^* = \mathbf{0}_{\mathbb{R}^n},$$

$$\partial_a L(\mathbf{v}^*, a^*; \phi) = -\langle \mathbf{y}, \phi \rangle = 0.$$

Substituting this expression of $\mathbf{v}^*$ in the definition of L, we get:

$$\begin{aligned}
L(\mathbf{v}^*, a^*; \phi) &= \frac{1}{2}\|\mathbf{v}^*\|^2 + \langle \phi, g(\mathbf{x}) \rangle + \iota_{\mathbb{R}_+^n}(\phi) \\
&= \frac{1}{2}\langle \mathbf{v}^*, \mathbf{v}^* \rangle + \langle \phi, g(\mathbf{x}) \rangle \\
&= \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} y_i y_j \phi_i \phi_j^* \langle \mathbf{x}_i^*, \mathbf{x}_j^* \rangle + \langle \phi, g(\mathbf{x}) \rangle \\
&= \frac{1}{2}\langle KD\phi, D\phi \rangle + \sum_{i=1}^{n} \phi_i \big(1 - y_i(\langle \mathbf{v}^*, \mathbf{x}_i \rangle + a^*)\big) \\
&= \frac{1}{2}\langle KD\phi, D\phi \rangle + \langle \phi, \mathbf{1} \rangle - \sum_{i=1}^{n} \phi_i y_i (\langle \mathbf{v}^*, \mathbf{x}_i \rangle + a^*) \\
&= \frac{1}{2}\langle KD\phi, D\phi \rangle + \langle \phi, \mathbf{1} \rangle - \sum_{i=1}^{n} \phi_i y_i \langle \mathbf{v}^*, \mathbf{x}_i \rangle - a^* \langle \phi, \mathbf{y} \rangle \\
&= \frac{1}{2}\langle KD\phi, D\phi \rangle + \langle \phi, \mathbf{1} \rangle - \langle KD\phi, D\phi \rangle \\
&= -\frac{1}{2}\langle KD\phi, D\phi \rangle + \langle \phi, \mathbf{1} \rangle,
\end{aligned}$$

and so the dual problem consists in maximizing $-\frac{1}{2}\langle KD\phi, D\phi \rangle + \langle \phi, \mathbf{1} \rangle$ under the constraints that $\phi_i \geq 0$ for all i and $\langle \phi, \mathbf{y} \rangle = 0$.

[3] That is, $\forall i \in [\![1, n]\!], \phi_i \geq 0$.

10. The classifier takes the following form, where a^* has been defined in the previous questions:

$$C : \mathbf{x} \mapsto \text{sign}(\langle \mathbf{v}^*, \mathbf{x} \rangle + a^*).$$

11. The only necessary data is the support vectors, that is the vectors $\mathbf{x}_i$ such that $\phi_i > 0$.